Chapter-2

Non-aqueous Solvents

TOPICS

- Some Definition
- Properties of Solvent
- Liquid ammonia
- Liquid hydrogen fluoride
- Sulfuric acid
- Fluorosulfonic acid
- Bromine trifluoride
- Dinitrogen tetraoxide
- Ionic liquids
- Supercritical fluids

Some Definition

- **A solution is a homogeneous mixture of a solute and a solvent.**
- **A solute is the material present in the smaller amount in the solution.**
- **A solvent is the material present in the larger amount in the solution and is responsible for dissolving the solute.**

Properties of Solvent

- **1. The temperature range over which it is a liquid.**
- **2. Its dielectric constant**
- **3. Its donor and acceptor properties**
- **4. Its protonic acidity and bacisity**
- **5. The nature and extent of autodissociation**
- **6. Hydrogen bonding**

Aqueous and Non-aqueous Solvents

- **Aqueous solutions contain liquid water as their solvent. Water molecules encircle and integrate the solute (dissolved) ions and molecules into the water's network of links. Following that, the dissolved species get diffused throughout the water.**

- **A non-aqueous solvent is not an organic chemical and is not water. Liquid ammonia, antimony trichloride, sulfuryl chloride, hydrogen fluoride, nitrogen tetroxide, pure sulphuric acid, sulfuryl chloride fluoride, phosphoryl chloride, bromine pentafluoride, and other inorganic acids are some of the most common examples of non-aqueous solvents.**

Types of Non-aqueous Solvents

1. **Aprotic Solvents:** These solvents act as catalysts and are chemically inert. These solvents can accelerate or reduce the reaction rate during the chemical process or reaction, depending on the situation. The most common aprotic solvents are chloroform and benzene, neither are acidic nor basic.
2. **Protogenic Solvents:** These are naturally acidic. They can donate a proton and improve the strength of weak bases. Hydrochloric acid and sulphuric acid are examples of protogenic solvents.
3. **Protophilic Solvents:** Natural photophilic solvents are basic and have a strong proton affinity. Liquid ammonia, amines, and ketone are examples of protophilic solvents.
4. **Amphiprotic Solvents:** These are protogenic and protophilic in nature. Amphiprotic solvents have acidic and basic properties. They also take the proton and donate it—for example, water, alcohols, and weak organic acids.

Aqueous Solutions

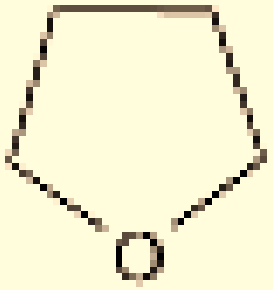
- Aqueous is a term used to describe a water-based system. Aqueous can also refer to a solution or mixture including **water as the solvent.** The chemical bonding type of a material determines whether or not it will form an aqueous solution.

Categories of non-aqueous solvents

- Protic solvents (e.g. HF, H_2SO_4, MeOH);
- Aprotic solvents (e.g. N_2O_4, BrF_3);
- Coordinating solvents (e.g. MeCN, Et_2O, Me_2CO).



diethyl ether



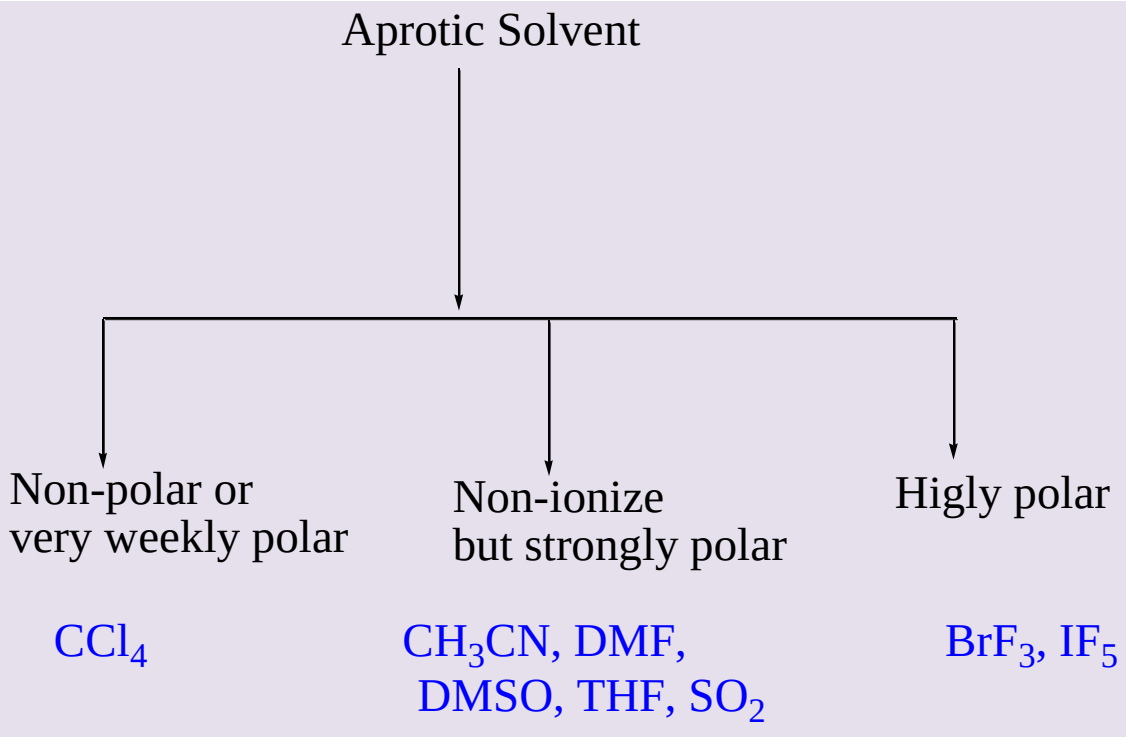
tetrahydrofuran (THF)



diglyme

Types of Solvent

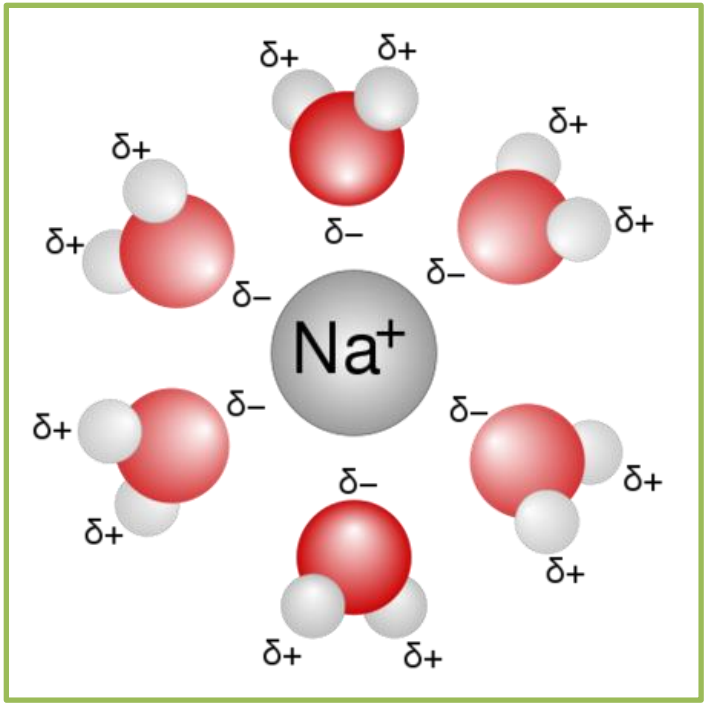
- Protic Solvent : H_2O, HCl, HF, H_2SO_4
- Aprotic Solvent :

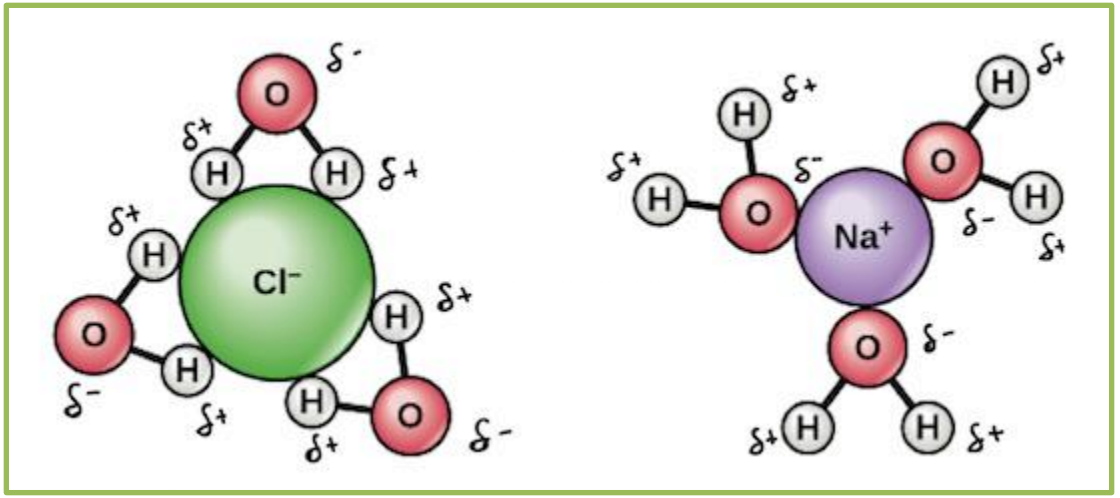


Types of Aqueous Solutions

1. **Electrolyte Solutions: Solutions that conduct electricity are known as electrolyte solutions. An electrolyte is a salt or ionisable molecule that permits that solution to conduct electricity when dissolved in a solution.**
2. **Strong and Weak Electrolytes: An electrolyte is a salt or ionisable molecule that permits a solution to conduct electricity. A strong electrolyte dissociates into its constituent ions in solution, whereas a weak electrolyte remains fundamentally undissociated. Weak electrolytes include acetic acid, which is also a weak acid.**
3. **Nonelectrolyte Solutions: Nonelectrolytes are substances that do not ionise in water. Glucose or $C_6H_{12}O_6$ is an example of a nonelectrolyte. Although glucose (sugar) dissolves quickly in water, it is classified as a nonelectrolyte because it does not break into ions in the solution and does not conduct electricity.**

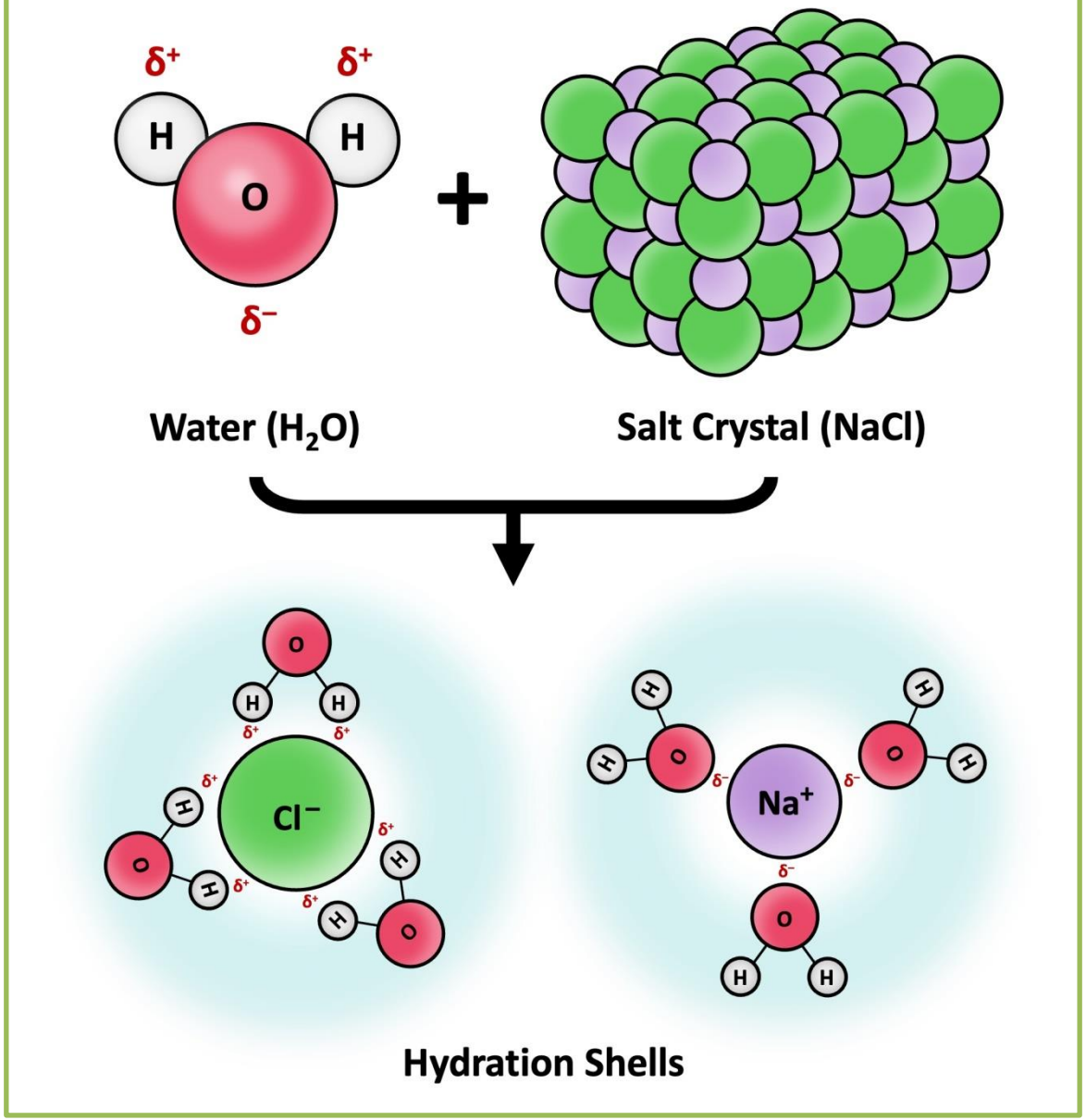
How Water Dissolves Substances?



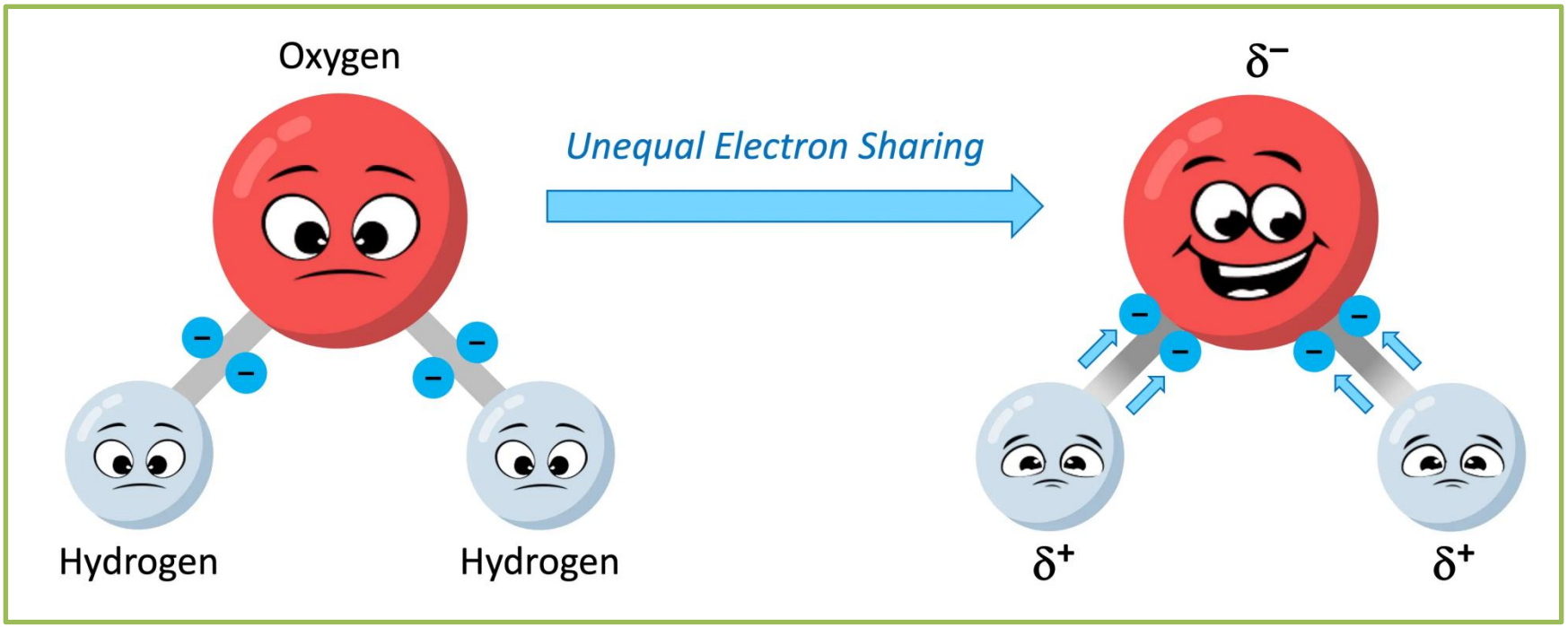


A sodium ion surrounded by water molecules. The positivity charged hydrogen ions are labeled as delta+ (shown in white) and the negatively charged oxygen atoms are labeled as delta- (shown in red).

Solvation

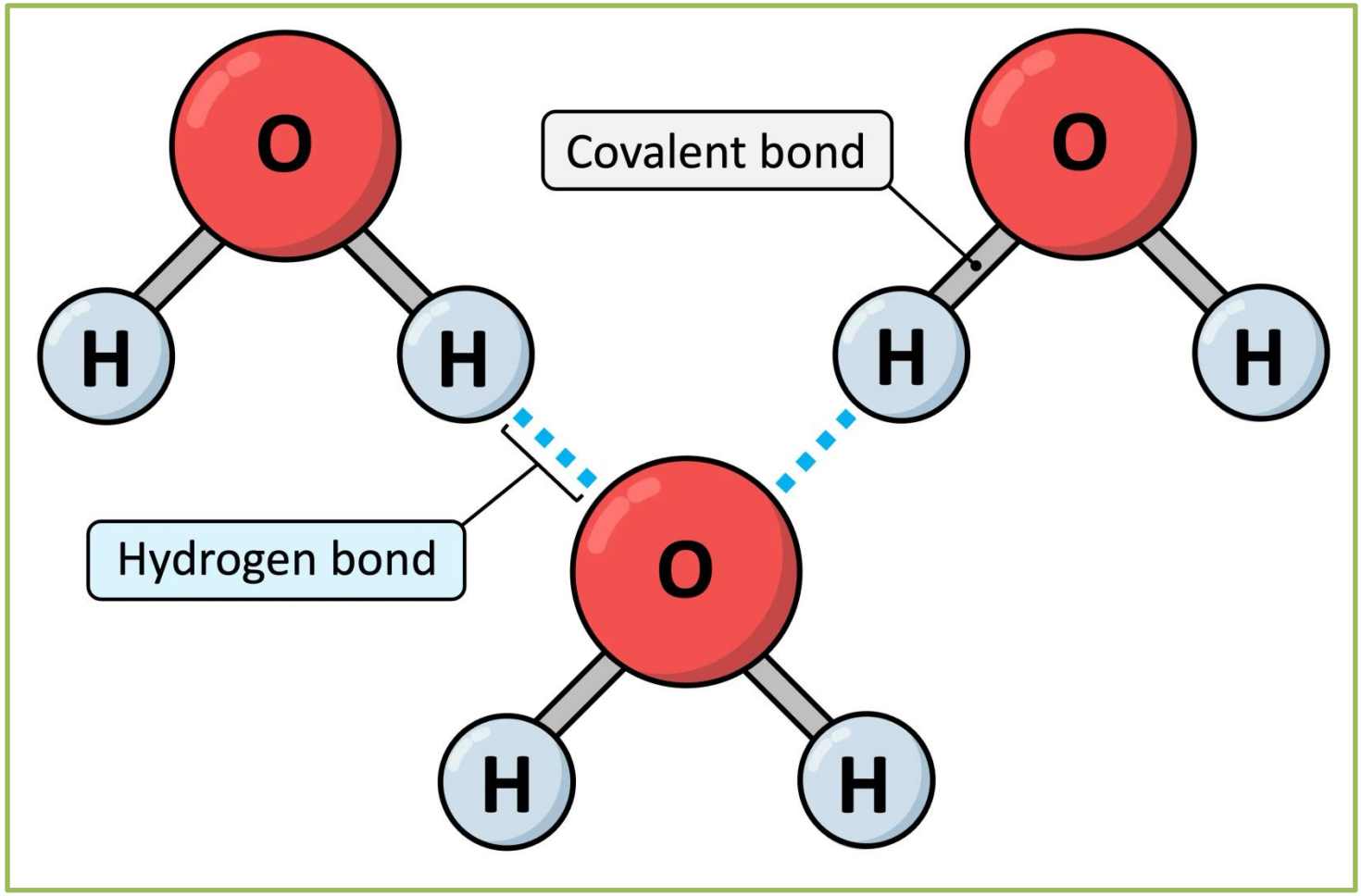


Polarity of Water



The slightly negative poles (δ^-) will attract the slightly positive poles (δ^+) of other molecules, and vice versa

Hydrogen Bonding



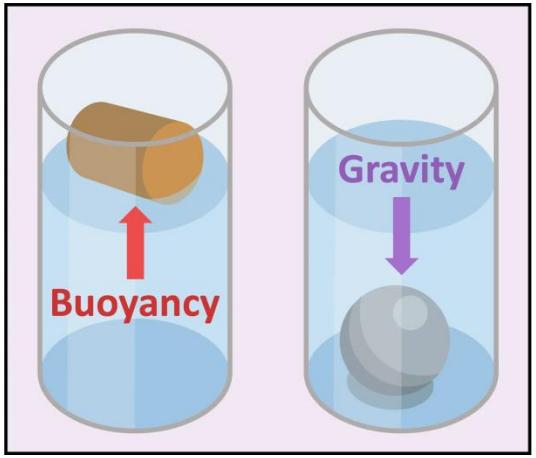
Hydrogen bonds are stronger than standard polar associations due to the particularly high electronegativity of F, O and N

Cohesive and Adhesive

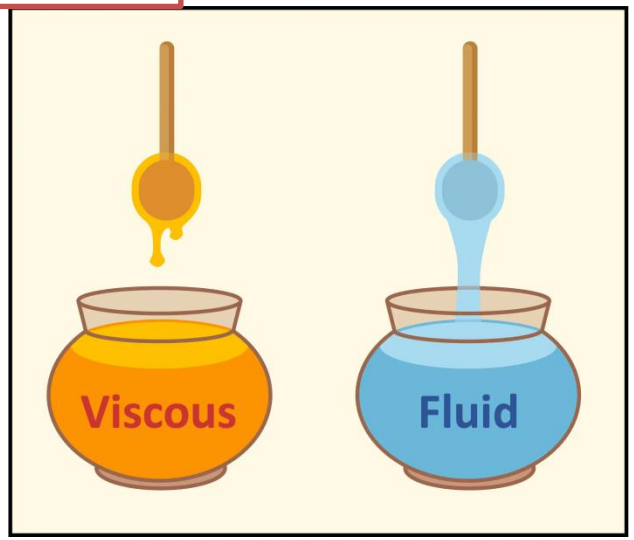
- Water is **cohesive** – a water molecule is able to stick to other water molecules via the formation of hydrogen bond.

- Water is also **adhesive** – a water molecule is able to stick to other polar or charged molecules via the formation of polar associations

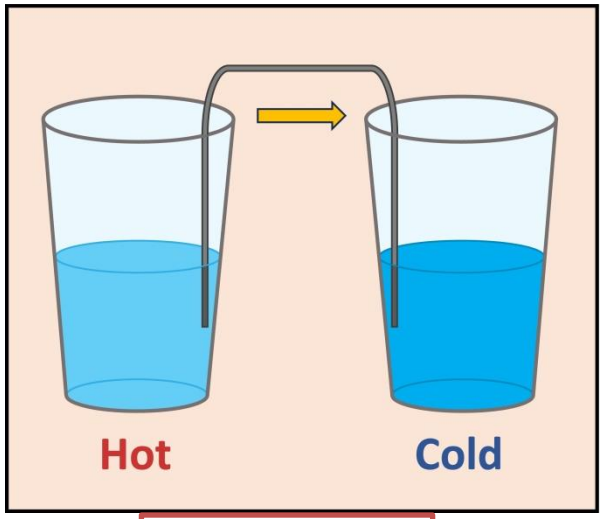
Physical Properties of Water



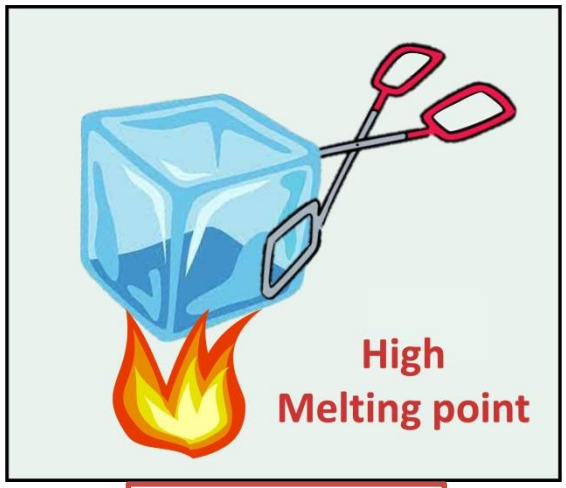
Buoyancy



Viscosity



Conductivity



Heat Capacity

Solution Vs colloid

- A solution is a homogeneous mixture where the particles are molecule-sized and cannot be separated by filtration. Light passes through the solution without scattering from the solute particles.

- A colloid, on the other hand, has particles that are larger than those in a solution (1-1000 nm) and can scatter light (Tyndall effect). The particles in a colloid do not settle out, making it appear to be cloudy but uniform and homogeneous.

Solution VS Suspension:

- **Solution**: A solution is a homogeneous mixture where the particles are very small, typically less
- **Suspension**: A suspension is a heterogeneous mixture where the particles are much larger, generally over 1,000 nanometers in size. than 1 nanometer in size.

Emulsion VS Suspension

In a **suspension**, two substances of any phase of matter, such as solid, liquid, or gas, can be found.

An **emulsion**, on the other hand, consists of only two immiscible liquids.

Suspensions examples include sand in water,

Emulsion examples include butter, margarine, and mayonnaise.

What is the levelling effect in water?

- The leveling effect of water is exerted on all strong acids and bases because all strong acids and bases are equally strong in the water. This is because all strong acids and bases dissociate entirely to form H_3O^+ and OH^- (respectively) ions when dissolved in water

The leveling effect of liquid ammonia

- Acids ionize in Liq.NH_3 to create NH_4^+ cations, liquid ammonia (NH_3) works as a leveling solvent for HCl (Strong acid) and CH_3COOH (Weak acid).
- $HCl + NH_3 \rightarrow Cl^- + NH_4^+$
- $CH_3COOH + NH_3 \rightarrow CH_3COO^- + NH_4^+$
-

Differentiating effect

- Differentiating solvent is the solvent in which distinct acids dissociate to varying degrees and so have different strengths. Weak bases serve as acid-specific solvents. It is the solvent's ability to distinguish between the strength of acids and bases:
- Examples:
- Water is a distinguishing solvent for amines (weak bases), which are far weaker than mineral alkalies (NaOH & KOH).
- When $HClO_4$, HCl, and HNO_3 are dissolved in glacial acetic acid, only $HClO_4$ is the strong acid, indicating that acetic acid distinguishes between the strength of mineral acids.

$$CH_3COOH + HClO_4 \rightleftharpoons CH_3COOH_2^+ + ClO_4^-$$

$$CH_3COOH + HBr \rightleftharpoons CH_3COOH_2^+ + Br^-$$

Types of solvent on the basis of proton interaction

- On the basis of proton interaction, solvents are of **four types**,
- (i) **Protophilic solvents**: Solvents which have greater tendency to accept protons, i.e., **water, alcohol, liquid ammonia**, etc.
- **(ii) Protogenic solvents**: Solvents which have the tendency to produce protons, i.e., water, liquid hydrogen chloride, glacial acetic acid, etc.
- **(iii) Amphiprotic solvents**: Solvents which act both as **protophilic or protogenic, e.g., water, ammonia, ethyl alcohol, etc.**
- **(iv) Aprotic solvents**: Solvents which neither donate nor accept protons, e.g., **benzene, carbon tetrachloride, carbon disulphide, etc.**

- **HCl** acts as an acid in H_2O, a stronger acid in NH_3, a weak acid in CH_3COOH, neutral in C_6H_6 and a weak base in HF.

- **The main difference between leveling solvent and differentiating solvent lies in the way they affect the strength of acids and bases. Here is a table comparing the two types of solvents:**

ability to enhance

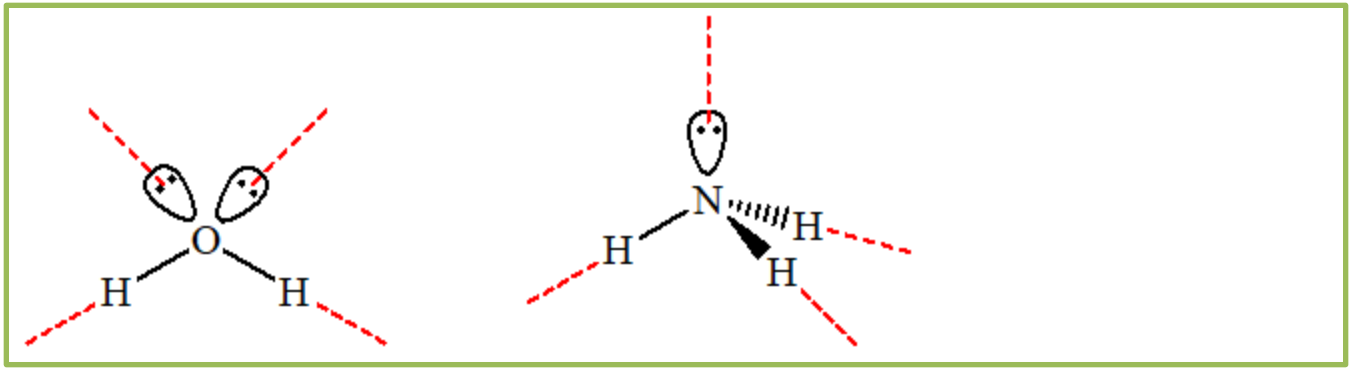
ability to distinguish

PROPERTY	LEVELING SOLVENT	DIFFERENTIATING SOLVENT
❖ Definition	A leveling solvent has the ability to enhance the strength of weak acids or bases to that of strong ones.	A differentiating solvent allows various acids to dissociate to different degrees, resulting in different strengths.
❖ Effect on Acids	In a leveling solvent, several acids are completely dissociated and are thus of the same strength.	In a differentiating solvent, various acids dissociate to different degrees and thus have different strengths.
❖ Effect on Bases	The strength of a strong base is leveled by the acidity of the solvent.	Strong bases are leveling solvents for acids, while weak bases are differentiating solvents for acids.
❖ Examples	Strong acids dissolved in water exhibit more strongly acidic properties than in other solvents due to the leveling effect.	Water is a differentiating solvent for acetic acid (CH_3COOH) because it is a weaker acid than hydronium ions (H_3O^+), but it exhibits a leveling effect on hydrochloric acid (HCl).

- An **inorganic nonaqueous solvent** is a solvent other than water, that is not an organic compound.

- Ammonia, hydrogen fluoride, sulfuric acid, as well as more specialized solvents, hydrazine, and selenium oxychloride etc.

Why is the boiling point of water higher than ammonia?



Each water molecule has 2 lone pairs of electrons and 2 hydrogen atoms to form hydrogen bonds. Averagely, there are 2 hydrogen bonds per water molecule.

Each ammonia molecule has 1 lone pair of electrons and 3 hydrogen atoms to form hydrogen bonds (hydrogen atoms is in excess). Averagely, there is 1 hydrogen bond per ammonia molecule.

Since water has more hydrogen bonds per molecule, more energy is needed to separate water molecules than ammonia molecules. Hence, the boiling point of water is higher than that of ammonia.

Comparison of Water and Ammonia

Chemical Formula	H_2O	NH_3
Molecular Structure	**Bent molecular shape due to two lone pairs on oxygen.-**	Pyramidal molecular shape due to one lone pair on nitrogen
Polarity	**Polar molecule due to the electronegativity difference between oxygen and hydrogen.-**	Polar molecule due to the electronegativity difference between nitrogen and hydrogen
Boiling Point:-	**Water: Boils at 100°C.-**	Ammonia: Boils at -33°C
Melting Point:-	**Water: Melts at 0°C**	Ammonia: Melts at -77.7°C
Density	**Water: Density of 1 g/cm3 at 4°C.-**	Ammonia: Density of 0.681 g/cm3 at -33°C
Solubility	**Water: Excellent solvent for polar and ionic compounds.-**	Ammonia: Good solvent for organic compounds and some inorganic compounds
Acidity/Basicity	**Water: Neutral pH of 7**	Ammonia: Weak base, can accept a proton to form NH4+

- (i) Both liquid ammonia and water are highly associated through strong hydrogen bonding.
  (ii) Both ammonia and water are good ionising solvents.
- (iii) The ionisation of ammonia and water is given as Both OH^- and NH_2^- are strong bases, eg: NaOH and $NaNH_2$.

Levelling and differentiating effects

- Non-aqueous solvents that are good proton acceptors (e.g.NH_3) encourage acids to ionize in them.
- In a basic solvent, all acids are strong.
- The solvent is said to exhibit *a levelling effect* on the acid, since the strength of the dissolved acid cannot exceed that of the protonated solvent.
- In aqueous solution, no acidic species can exist that is a stronger acid than H_3O^+.
- Note, because of the *common ion effect*, if HCl is dissolved in acetic acid, the extent of ionization is less than in water and HCl acts as a weak acid.

Hydrogen bromide and iodide behave similarly but the extent of ionization of the three hydrogen halides varies along the series: HI > HBr > HCl.

This contrasts with the fact that all three compounds are classed as strong acids (i.e. fully ionized) in aqueous solution.

Thus, acetic acid exerts *a differentiating effect* on the acidic behaviour of HCl, HBr and HI, whereas water does not.

LIQUID AMMONIA as an non-aqueous solvent

- FREEZING POINT:- -77.7 ċ
- BOILING POINT:- -33.38 ċ
- DIELECTRIC CONSTANT:- 22.0 at -33.5 ċ
- LIQUID RANGE:- -77 to -33 ċ
- HEAT OF FUSION:- 0.018 kj mol^{-1}
- HEAT OF VAPORISATION:- 23.6 kj mol^{-1}
- SELF IONISABLE IN NATURE
- ACTS AS AN ASSOCIATED SOLVENT

WHY AMMONIA ACTS AS A BETTER SOLVENT THAN WATER:-

1. POOR CONDUCTOR OF ELECTRICITY
2. SPECIFIC HEAT OF AMMONIA IS GREATER THAN WATER
3. LESS VISCOUS THAN WATER
4. HIGH CRITICAL TEMPERATURE AND PRESSURE
5. LESS ASSOCIATED THAN WATER(DUE TO LESSER HYDROGEN BONDING)
6. DUE TO FORMATION OF STRONG REDUCING METAL –AMMONIA SOLUTIONS WITH AIKALI METALS.

Reactions in Liquid NH_3

1. Acid Base Reaction.
2. Precipitation Reaction.
3. Redox Reaction.
4. Solvation Reaction.
5. Metatheses Reactions

1. Acid Base Reaction.

1. $NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH^-$
2. "This type of reaction is an acid-base reaction in terms of Bronsted Lowry protonic concept ."
3. The following reaction is an acid-base reaction in terms of Bronsted Lowry protonic concept:

4. NH_3Acid(I)+NH_3Base(II)$\rightleftharpoons$$NH_2^-$Base(I) + NH_4+Acid(II)
5. Here, NH_3 acts as a proton donor as well as proton acceptor.

- **NEUTRALISATION OF AN ACID AND BASE IN LIQUID AMMONIA INVOLVES COMBINATION OF NH_4^+ ION(FROM ACID) AND NH_2^-**
- **ION(FROM BASE):- $NH_4Cl + KNH_2 \rightarrow KCl + 2NH_3$**

  ACID BASE

  $NH_4^+ + NH_2^- \rightarrow 2NH_3$
- **AS AN ACID:- $Zn(OH)_2 + KOH \rightarrow K_2Zn(NH_2)_4$**
- **AS A BASE:- $Zn(NH_2)_2 + 2NH_4Cl \rightarrow ZnCl_2 + 4NH_3$**

2. Precipitation Reaction

1. **PRECIPITATION REACTIONS INVOLVE DOUBLE DECOMPOSITION BECAUSE OF THE DIFFERENCES IN SOLUBILITIES.**
2. SILVER CHLORIDE PRECIPITATED IN AQUEOUS SOLUTION AS:-

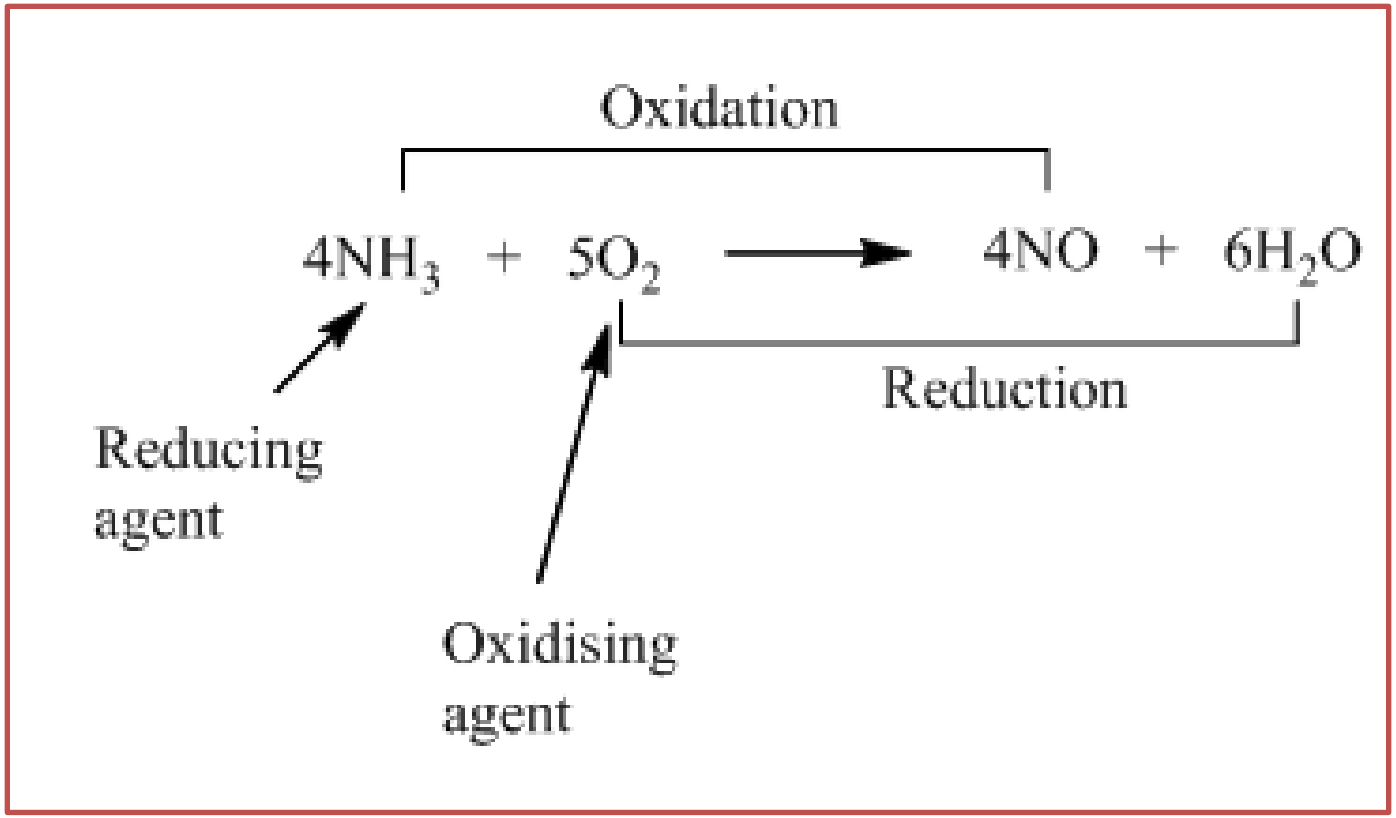


3. WHITE PPT OF $BaCl_2$ IS PRODUCED WHEN SILVER CHLORIDE AND LIQ. AMMONIA BROUGHT TOGETHER :-



3. Redox Reaction

- For **$N_2 + H_2 = NH_3$** the oxidation numbers on the N and the H change going from reactants to products.
- Therefore $N_2 + H_2 = NH_3$ is a redox reaction.
- The reactant that releases electrons is oxidized and it will be a reducing agent. The reactant that receives electrons is reduced and it will be an oxidizing agent.


Oxidation
4NH3 + 5O2
4NO + 6H2O
Reduction
Reducing agent
Oxidising agent

4. Solvation Reaction.

- THE SOLVOLYSIS REACTIONS IN LIQUID AMMONIA ARE CALLED AMMONOLYSIS OR AMMONOLYTIC REACTIONS.
- HYDROLYSIS OF $SiCl_4$ OCCURS AS:-
- $SiCl_4 + 4H_2O \leftrightarrow Si(OH)_4$
- (Silicic acid)
- ORGANIC HALIDES UNDERGO SLOW AMMONOLYSIS REACTION TO FORM AMMINES :-

$RX + 2NH_3 \rightarrow RNH_2 + NH_4X$

(Primary amines)

5. Metatheses Reactions

- A metathesis reaction, sometimes called a double replacement reaction, double displacement reaction, or double decomposition reaction, is a chemical process involving the exchange of bonds between two non-reacting chemical species which results in the creation of products with similar or identical bonding.

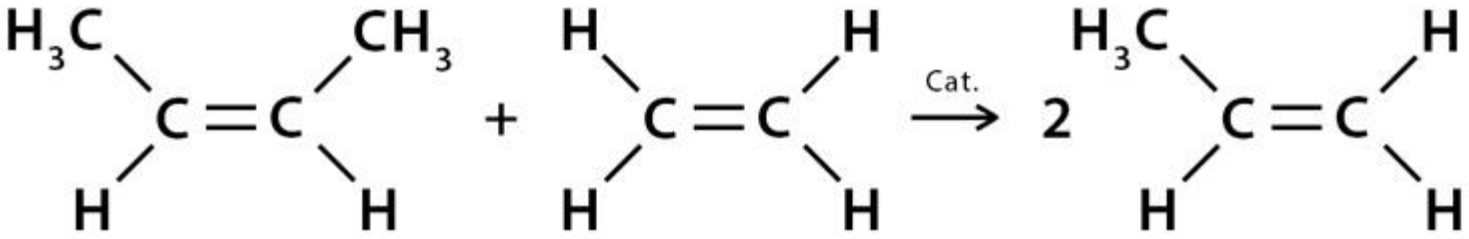


- $AgCl + KNO_3 \rightarrow KCl(s)\ (ppt) + AgNO_3$

- $Ba(NO_3)_2 + 2AgCl \rightarrow BaCl_2(ppt) + 2AgNO_3$

Metathesis Examples

1. Olefin Metathesis



2-Butene + Ethene → 2 Propene

2. Salt Metathesis

$AgNO_3$ (aq.) + NaCl (aq.)

Silver nitrate + Sodium chloride

$NaNO_3$ (aq.) + AgCl (aq.) ↓

Sodium nitrate + Silver chloride

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SOLUTIONS IN LIQUID AMMONIA

- **THE MOST STRIKING PROPERTY OF LIQUID AMMMONIA IS ITS ABILITY TO DISSOLVE ALKALI METALS.THE RESULTING SOLUTIONS ARE BLUE AND GOOD ELECTRICAL CONDUCTORS.**
- **WHEN ALKALI METALS ARE DISSOLVED IN LIQUID AMMONIA THEY IONISE TO GIVE METAL IONS AND VALENCE ELECTRONS AS:-**
- **$Na \leftrightarrow Na^{+} + e^{-}$**
- **Both alkali metal and electron become solvated by ammonia molecules.**

- $Na^+ + xNH_3 \rightarrow [\ Na(NH_3)x]^+$

  (ammoniated cation)
- $e^- + yNH_3 \rightarrow [\ e(NH_3)y]^-$

  (amnoniated electron)
- THE COMPLETE REACTION MAY BE WRITTEN AS:- $Na \rightarrow [\ Na^+ (NH_3)x]^+ + [\ e\ (NH_3)y]^-$
- THE AMMONIATED ELECTRONS ARE RESPONSIBLE FOR BLUE COLOUR OF SOLUTION.

Alkali Metals in Liquid Ammonia

- Alkali metals (e.g., Na) in NH3(*l) produce bronze colored solutions when*
- concentrated (> 1.0 M) and blue solutions when less concentrated.
- • In both cases, the solution contains solvated electrons.

$$M(s) + NH_3(l) \longrightarrow [M(NH_3)^+] + e^- (solvated)$$

- The more dilute blue solutions are strongly paramagnetic, becoming less so with
- increasing concentration.
- • In the concentrated solutions, extensive electron pairing occurs, making the
- bronze colored solutions less paramagnetic.
- • Solutions of metals in liquid ammonia are very strong reducing agents and react
- very much like the free metals themselves.
- • The reduction process appears to involve free solvated electrons.

Reduction of Ammonia by Alkali Metals

- The resistance of ammonia itself to reduction provides a medium in which highly
- reduced states are thermodynamically stable.
- • Alkali metals do react very slowly with NH3 in a manner analogous to their behavior with water.

$$2Na(s) + 2NH_3(l) \longrightarrow [2Na(NH_2)] + H_2$$

- The reaction is catalyzed by small amounts of any Fe(III) salt.
- • Catalytic mechanism is uncertain, but appears to involve rapid reduction of Fe(III)
- to produce finely divided Fe(0), which may be the actual catalytic agent.

Solutions of s-block metals in liquid NH_3

All Group 1 metals and group 2 metals Ca, Sr, and Ba dissolve in liquid NH_3.

Dilute solutions of the metals are bright blue, color arising from a broad and intense absorption in the IR region.

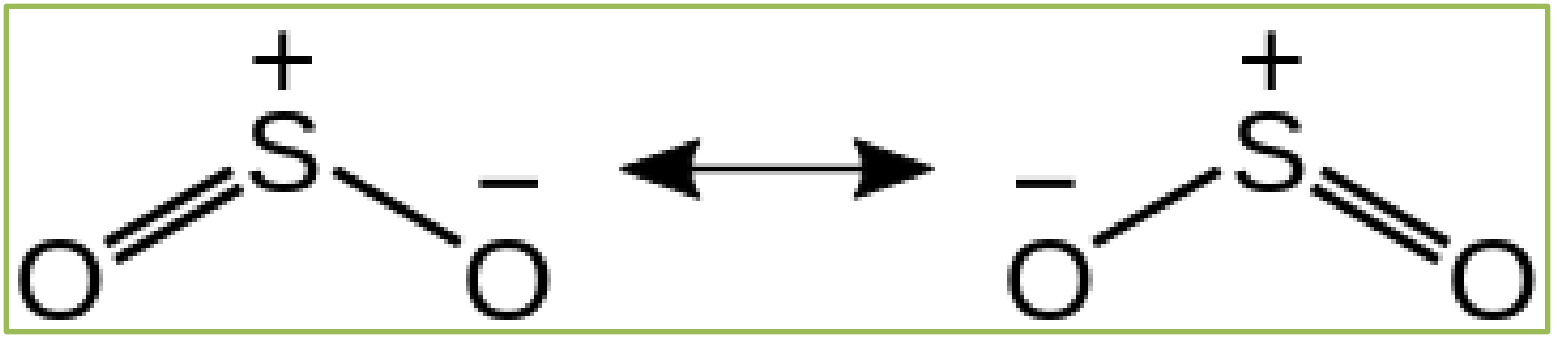
$$M \xrightarrow{\text{dissolve in liquid } NH_3} M^+(solv) + e^-(solv)$$

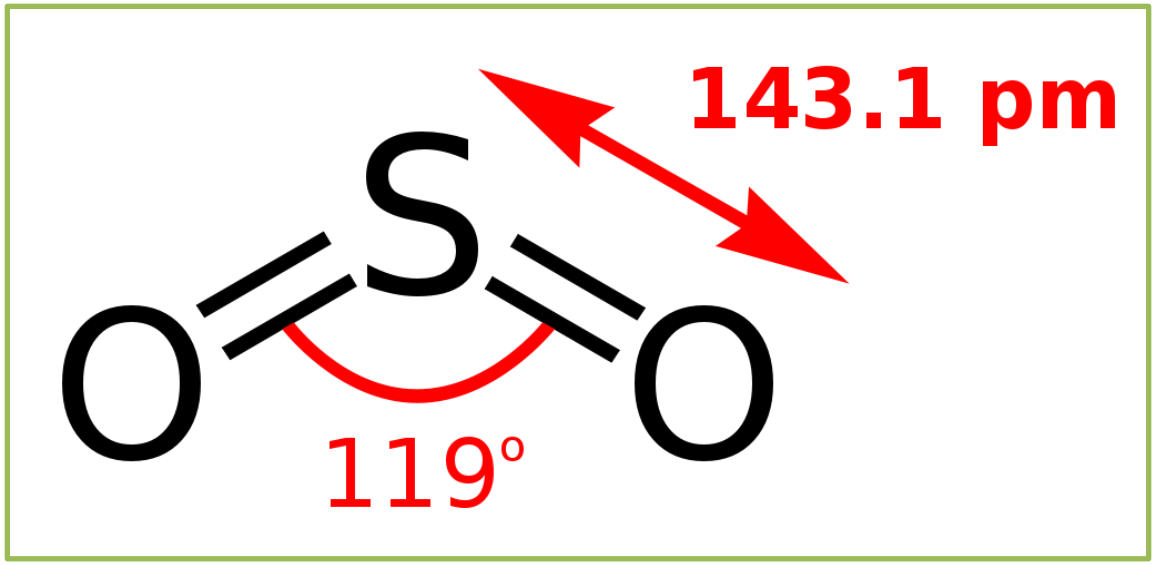
Dilute solutions occupy a volume greater than the sum of the metal plus solvent, with electrons occupying cavities of radius 300-400 pm.

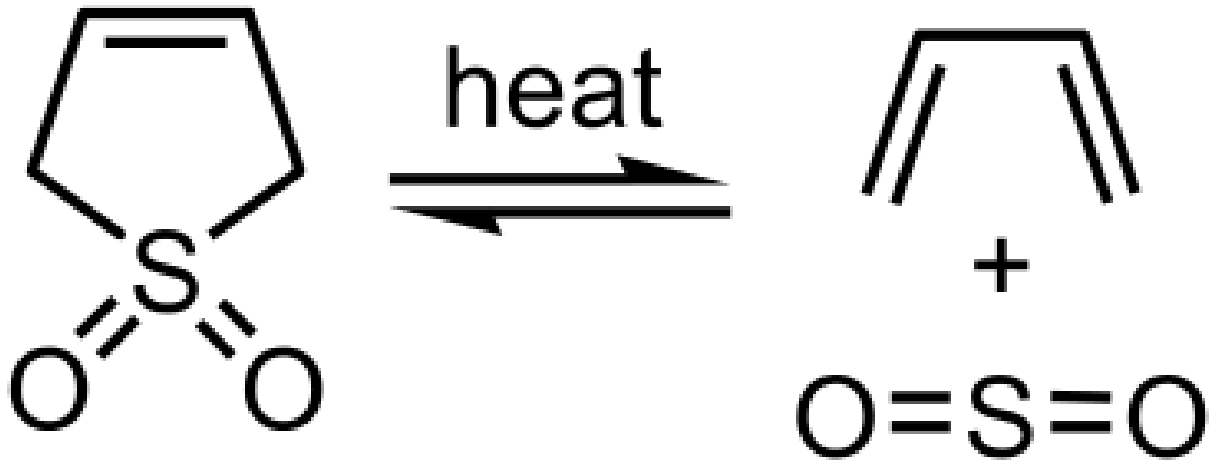
- Dilute solutions are paramagnetic and the magnetic response corresponds to that of one free electron per metal atom.
- Molar conductivity initially decreases with increasing concentration, reaching a minimum near 0.05 M. Conductivity increases at higher concentrations and in saturated solutions the concentration is comparable with the solid metal.
- Saturated solutions are not blue and paramagnetic, but instead are bronze and diamagnetic. They are essentially 'metal-like' and have been described as expanded metals.
- The blue solutions of alkali metals in liquid NH_3 decompose very slowly, liberating H_2 as the solvent is reduced. $2NH_3 + 2e^- \rightarrow 2[NH_2]^- + H_2$
- Used as a reducing agent in a variety of reactions

LIQUID SULPHUR DIOXIDE :-

- IT IS A NON –PROTONIC SOLVENT OR APROTIC SOLVENT BECAUSE IT DOES NOT CONTAIN ANY HYDROGEN ATOM.
- IT IS ALSPO ONE OF THE IMPORTANT NON AQUEOUS SOLVENT AND WIDELY USED IN INDUSTRY.
- PHYSICAL PROPERTIES OF LIQUID SO_2 :-
- FREEZING POINT :- -75.46 ċ
- BOILING POIN T :- -10.02 ċ
- DIELECTRIC CONSTANT:- 17.40


S
O
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S
O
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143.1 pm
S
O
O
119°


heat
S
O
O
+
O=S=O

REACTIONS IN LIQUID SO_2 :-

1) ACID BASE RACTION :-

Liquid SO_2 undergo auto ionisation as :

$$2SO_2 \leftrightarrow SO^{2+} + SO_3^{2-}$$

Acid Base or Neutralisation reaction in liquid SO_2 may be given as:-

$$SOCl_2 + Cs_2SO_3 \rightarrow 2CsCl + 2SO_2$$

(ACID) (BASE)

2) SOLVOLYTIC REACTIONS :-

- Solvolytic reactions in liquid SO_2 are not so common as hey are in other solvents such as NH_3.
- Certain covalent halides such as PCl_5 ,PBr_5, WCl_6 under go solvolysis in liquid SO_2 in sealed tubes.
- $PCl_5 + SO_2\ (liq) \rightarrow POCl_3 + SOCl_2$
- $PBr_5 + SO_2(liq) \rightarrow POBr_3 + SOBr_2$
- $WCl_6 + SO_2\ (liq) \rightarrow WOCl_4 + SOCl_2$

3) PRECIPITATION REACTIONS :-

- Precipitation of several insoluble compounds in liq. Sulphur dioxide can be carried out by treating with soluble compound in SO_2.

$$SbCl_3 + 3LiI \rightarrow SbI_3 \downarrow + 3LiCl$$

$$AlCl_3 + 3NaI \rightarrow 3NaCl \downarrow + AlI_3$$

4) REDOX REACTIONS :-

- Liq. Sulphur dioxide does show any marked reducing or oxidising property. It simply act as a medium for certain redox reactions.

$$6KI + 3SbCl_5 \rightarrow 2K_3[SbCl_6] + SbCl_3 + 3I_2$$

Liquid hydrogen fluoride

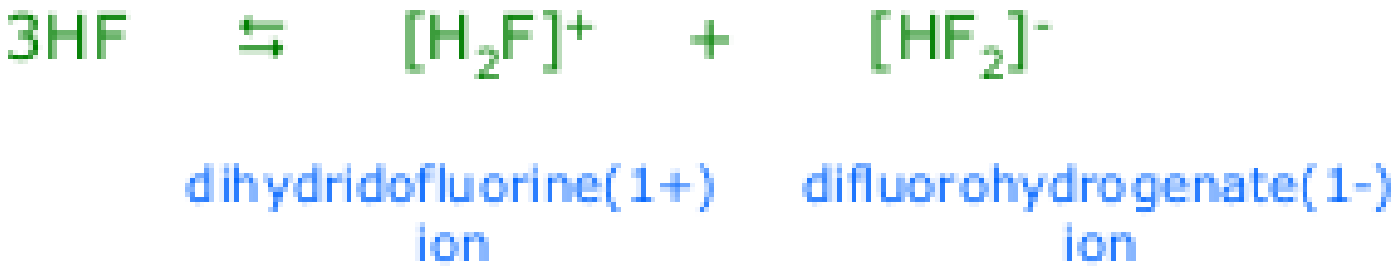
Physical properties

Hydrogen fluoride attacks silica glass thereby corroding glass reaction vessel)and it is only relatively recently that HF has found applications as a non-aqueous solvent. It can be handled in polytetrafluoroethene (PTFE) containers, or, if absolutely free of water, in Cu or Monel metal (a nickel alloy) equipment.



Liquid range: 190 to 292.5 K
Relative permittivity 84 at 273 K; 175 at 200 K

Liquid HF undergoes self-ionization: $K_{self} = 2 \times 10^{-12}$ at 273 K



$$MeCO_2H + 2HF \longrightarrow [MeC(OH)_2]^+ + [HF_2]^-$$

$$EF_5 + 2HF \rightleftharpoons [H_2F]^+ + [EF_6]^- \quad E = As \text{ or } Sb$$

$$BF_3 + 2HF \rightleftharpoons [H_2F]^+ + [BF_4]^-$$

$$PF_5 + 2HF \rightleftharpoons [H_2F]^+ + [PF_6]^-$$

$$BrF_3 + HF \rightleftharpoons [BrF_2]^+ + [HF_2]^-$$

Few protic acids are able to exhibit acidic behaviour in liquid HF, on account of the competition between HF and the solute as H^+ donors. Perchloric acid and fluorosulfonic acid do act as acids.

$$HOSO_2F + HF \rightleftharpoons [H_2F]^+ + [SO_3F]^-$$

With SbF_5, HF forms a superacid which is capable of protonating very weak bases including hydrocarbons

$$2HF + SbF_5 \rightleftharpoons [H_2F]^+ + [SbF_6]^-$$

9.8 Sulfuric acid and fluorosulfonic acid

Physical properties of H_2SO_4

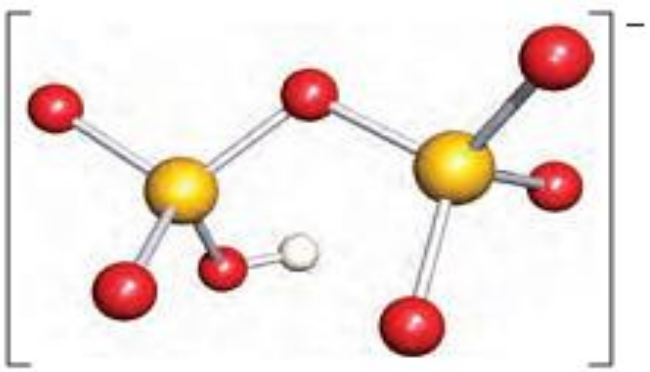
Table 9.6 Selected physical properties of sulfuric acid, H_2SO_4.

Property / units	Value
Melting point / K	283.4
Boiling point / K	≈603
Density of liquid / g cm^{-3}	1.84
Relative permittivity	110 (at 292 K)
Self-ionization constant	2.7×10^{-4} (at 298 K)

The value of the equilibrium constant for the self-ionization process 9.46 is notably large. In addition, other equilibria such as 9.47 are involved to a lesser extent.

$$2H_2SO_4 \leftrightarrows [H_3SO_4]^+ + [HSO_4]^- \qquad K_{self} = 2.7\times10^{-4} \qquad 9.46$$

$$2H_2SO_4 \leftrightarrows [H_3O]^+ + [HS_2O_7]^- \qquad K_{self} = 5.1\times10^{-5} \qquad 9.47$$



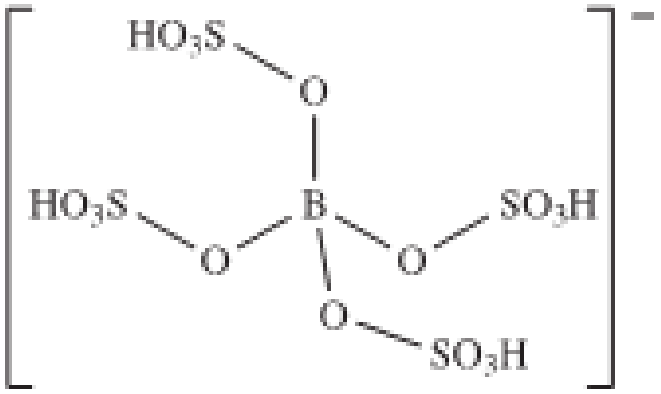
$[H_2S_2O_7]^-$

Acid - base behaviour in liquid H_2SO_4

$$MeCO_2H + H_2SO_4 \rightarrow [MeC(OH)_2]^+ + [HSO_4]^-$$

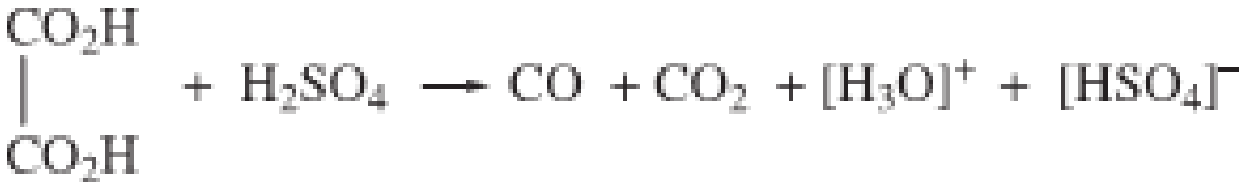
$$HNO_3 + 2H_2SO_4 \rightarrow [NO_2]^+ + [H_3O]^+ + 2[HSO_4]^-$$

$$H_3BO_3 + 6H_2SO_4 \rightarrow [B(HSO_4)_4]^- + 3[H_3O]^+ + 2[HSO_4]^-$$



$$H_3BO_3 + 2H_2SO_4 + 3SO_3 \rightarrow [H_3SO_4]^+ + [B(HSO_4)_4]^-$$

$$H[B(HSO_4)_4] + KHSO_4 \rightarrow K[B(HSO_4)_4] + H_2SO_4$$

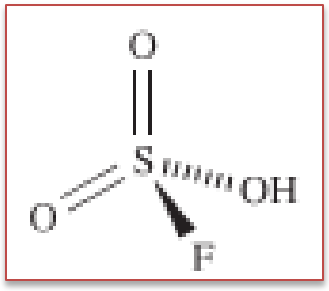


Physical properties of fluorosulfonic acid HSO_3F

Table 9.7 Selected physical properties of fluorosulfonic acid, HSO_3F.

Property / units	Value
Melting point / K	185.7
Boiling point / K	438.5
Density of liquid / g cm^{-3}	1.74
Relative permittivity	120 (at 298 K)
Self-ionization constant	4.0×10^{-8} (at 298 K)

It has a relatively long liquid range and a high dielectric constant.

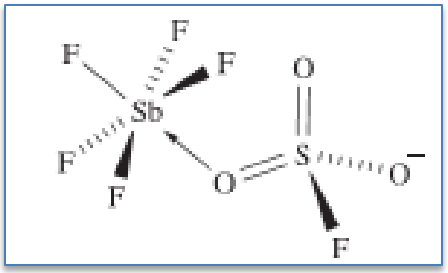


$$2HSO_3F \rightleftharpoons [H_2SO_3F]^+ + [SO_3F]^-$$

9.9 Superacids

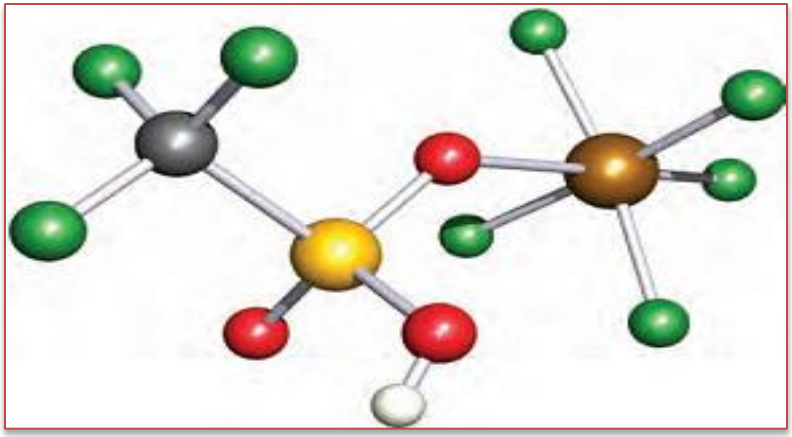
Superacids, capable of protonating even hydrocarbons, include mixtures of:

- HF and SbF_5
- HSO_3F and SbF_5 (called magic acid)



$$2HSO_3F + SbF_5 \rightleftharpoons [H_2SO_3F]^+ + [F_5SbOSO_2F]^-$$

The following figure shows the crystallographically determined structure of the related adduct $SbF_5OSO(OH)CF_3$



In superacidic media, hydrocarbons act as bases, and this is an important route to the formation of carbenium ions,

$$Me_3CH + \underbrace{[H_2SO_3F]^+}_{\text{generated in magic acid}} \longrightarrow [Me_3C]^+ + H_2 + HSO_3F$$

Carborane Superacids

Recently, a new class of superacids has been discovered: in contrast to well-established superacids, *carbaborane* (or *carborane*) *acids* possess chemically inert, extremely weak, conjugate bases. Carbaboranes are molecular clusters containing carbon and boron atoms, and the negative charge of a carbaborane monoanion is delocalized over the cluster.

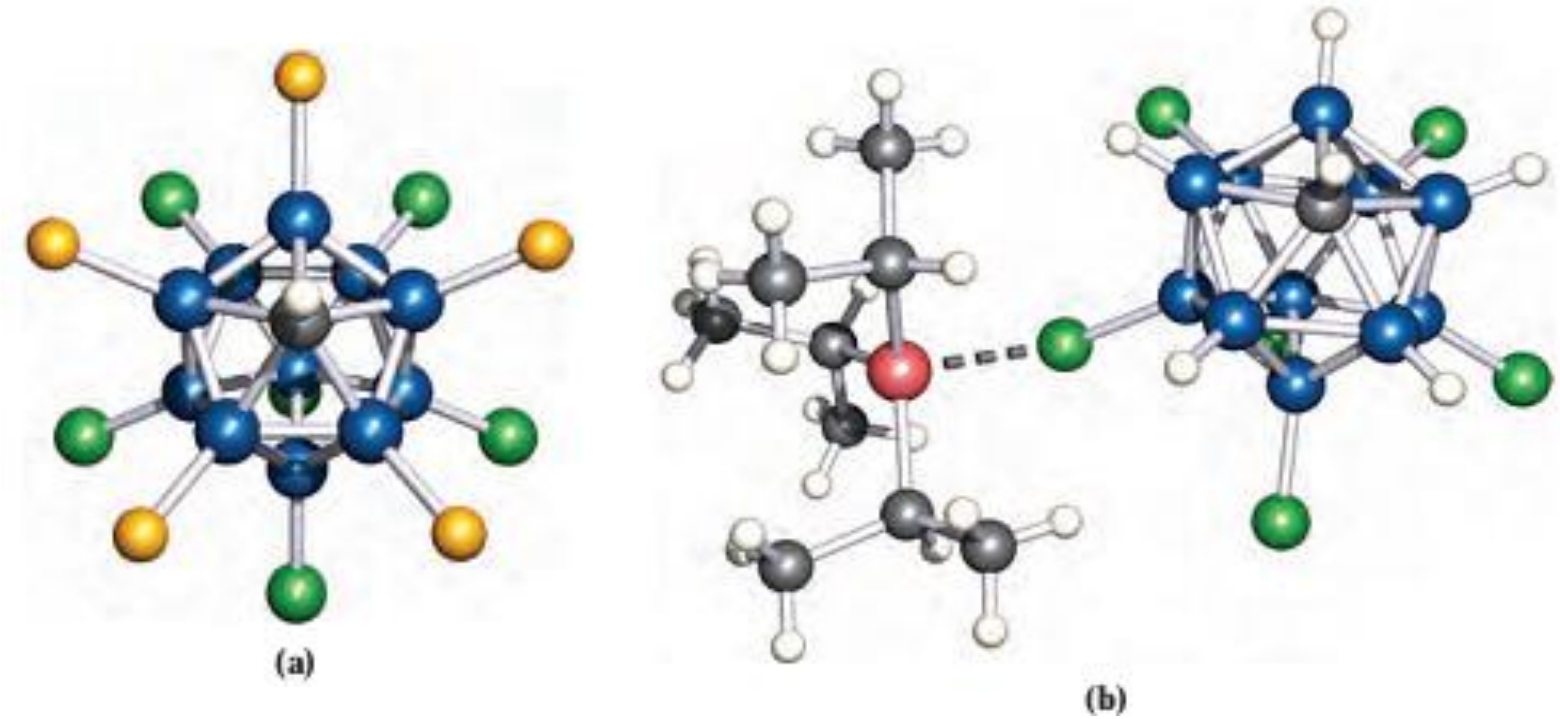


Fig. 9.6 (a) The $[CHB_{11}R_5X_6]^-$ (R = H, Me, Cl and X = Cl, Br, I) anions that are the conjugate bases of the family of carbaborane superacids. Colour code: C, grey; B, blue; H, white; R, yellow; X, green. (b) The structure of $^iPr_3Si(CHB_{11}H_5Cl_6)$ (X-ray diffraction) showing the long Si–Cl 'bond' which suggests that the structure approaches an ion-pair [Z. Xie *et al.* (1996) *J. Am. Chem. Soc.*, vol. 118, p. 2922]. Colour code: C, grey; B, blue; Cl, green; Si, pink; H, white.

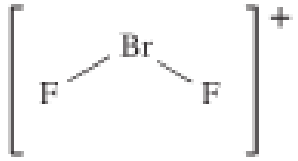
9.10 Bromine trifluoride (Good oxidizing and fluorinating agent)

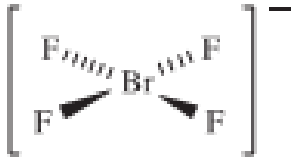
Physical properties

Table 9.8 Selected physical properties of bromine trifluoride, BrF_3.

Property / units	Value
Melting point / K	281.8
Boiling point / K	408
Density of liquid / g cm^{-3}	2.49
Relative permittivity	107
Self-ionization constant	8.0×10^{-3} (at 281.8 K)

$$2BrF_3 \rightleftharpoons [BrF_2]^+ + [BrF_4]^-$$





Behaviour of fluoride salts and molecular fluorides in BrF_3

Bromine trifluoride acts as a Lewis acid, readily accepting F^-. When dissolved in BrF_3, alkali metal fluorides, BaF_2 and AgF combine with the solvent to give salts containing the $[BrF_4]^-$ anion, e.g. K $[BrF_4]$, Ba$[BrF_4]_2$ and Ag$[BrF_4]$.

On the other hand, if the fluoride solute is a more powerful F^- acceptor than BrF_3, salts containing $[BrF_2]^+$ may be formed

$$KF + BrF_3 \longrightarrow K^+ + [BrF_4]^-$$

$$SbF_5 + BrF_3 \longrightarrow [BrF_2]^+ + [SbF_6]^-$$

$$SnF_4 + 2BrF_3 \longrightarrow 2[BrF_2]^+ + [SnF_6]^{2-}$$

$$AuF_3 + BrF_3 \longrightarrow [BrF_2]^+ + [AuF_4]^-$$

Conductometric measurements on solutions containing $[BrF_2][SbF_6]$ and $Ag[BrF_4]$,or $[BrF_2]_2[SnF_6]$ and $K[BrF_4]$ exhibit minima at 1 : 1 and 1 : 2 molar ratios of reactants respectively. These data support the formulation of neutralization reactions

$$\underset{\text{acid}}{[BrF_2][SbF_6]} + \underset{\text{base}}{Ag[BrF_4]} \longrightarrow Ag[SbF_6] + 2BrF_3$$

$$\underset{\text{acid}}{[BrF_2]_2[SnF_6]} + \underset{\text{base}}{2K[BrF_4]} \longrightarrow K_2[SnF_6] + 4BrF_3$$

Reactions in BrF_3

Much of the chemistry studied in BrF_3 media involves fluorination reactions, and the preparation of highly fluorinated species.

$$Ag + Sb \xrightarrow{\text{in } BrF_3} Ag[SbF_6]$$

$$2KCl + Sn \xrightarrow{\text{in } BrF_3} K_2[SnF_6]$$

$$Ag + Au \xrightarrow{\text{in } BrF_3} Ag[AuF_4]$$

$$KCl + VCl_4 \xrightarrow{\text{in } BrF_3} K[VF_6]$$

$$2ClNO + SnCl_4 \xrightarrow{\text{in } BrF_3} [NO]_2[SnF_6]$$

$$Ru + KCl \xrightarrow{\text{in } BrF_3} K[RuF_6]$$

Non-aqueous solvents that behave similarly to BrF_3 in that they are good oxidizing and fluorinating agents include ClF_3, BrF_5 and IF_5.

9.11 Dinitrogen tetraoxide (Good oxidizing and nitrating agent)

Physical properties

Table 9.9 Selected physical properties of dinitrogen tetraoxide, N_2O_4.

Property / units	Value
Melting point / K	261.8
Boiling point / K	294.2
Density of liquid / g cm^{-3}	1.49 (at 273 K)
Relative permittivity	2.42 (at 291 K)

The proposed self-ionization process for N_2O_4 is given in equation:

$$N_2O_4 \rightleftharpoons [NO]^+ + [NO_3]^-$$

Reactions in N_2O_4

Electropositive metals such as **Li and Na** react in liquid N_2O_4 liberating NO

$$Li + N_2O_4 \longrightarrow LiNO_3 + NO$$

Less reactive metals may react rapidly if ClNO, $[Et_4N][NO_3]$ or an organic donor such as MeCN is present. These observations can be explained as follows.

- ClNO can be considered to be a very weak acid in liquid N_2O_4, and hence encourages reaction with metals (equation 9.76).

$$Sn + 2ClNO \xrightarrow{\text{in liquid } N_2O_4} SnCl_2 + 2NO \qquad (9.76)$$

- $[Et_4N][NO_3]$ functions as a base in liquid N_2O_4 and its action on metals such as Zn and Al arises from the formation of nitrato complexes (equation 9.77) analogous to hydroxo complexes in an aqueous system. Figure 9.7 shows the structure of $[Zn(NO_3)_4]^{2-}$.

$$Zn + 2[Et_4N][NO_3] + 2N_2O_4 \longrightarrow [Et_4N]_2[Zn(NO_3)_4] + 2NO \qquad (9.77)$$

- Organic donor molecules appear to facilitate reactions with metals by increasing the degree of self-ionization of the solvent as a result of adduct formation with the $[NO]^+$ cation; e.g. Cu dissolves in liquid N_2O_4/MeCN

$$Cu + 3N_2O_4 \xrightarrow{\text{in presence of MeCN}} [NO][Cu(NO_3)_3] + 2NO$$

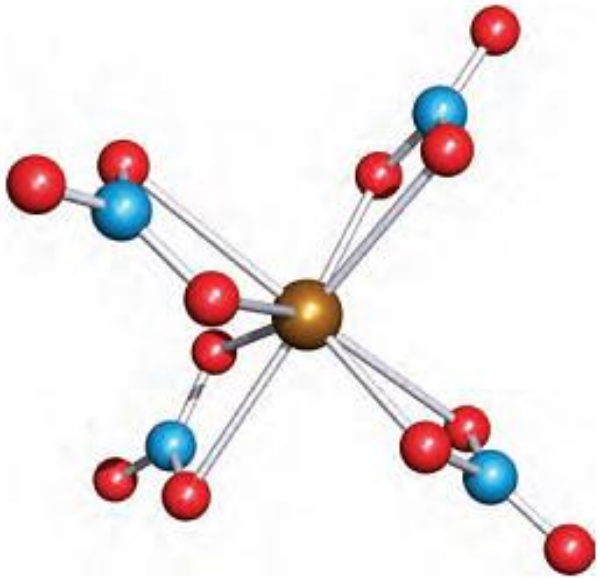


Fig. 9.7 $[Zn(NO_3)_4]^{2-}$

9.12 Ionic liquids (also called *molten* or *fused salts*)

A ***eutectic*** is a mixture of two substances and is characterized by a sharp melting point lower than that of either of the components; a eutectic behaves as though it were a single substance.

For example, the melting point of NaCl is 1073 K, but is lowered if $CaCl_2$ is added as in the Downs process.

In the solid state, $HgCl_2$ forms a molecular lattice, and layer structures are adopted by $HgBr_2$ (distorted CdI_2 lattice) and HgI_2.

$$2HgBr_2 \rightleftharpoons [HgBr]^+ + [HgBr_3]^-$$

An important group of molten salts with more convenient operating temperatures contain the tetrachloroaluminate ion, $[AlCl_4]^-$; an example is an $NaCl–Al_2Cl_6$ mixture. The melting point of Al_2Cl_6 is 463 K (at 2.5 bar), and its addition to NaCl (melting point, 1073 K) results in a 1:1 medium with a melting point of 446 K.

$$Al_2Cl_6 + 2Cl^- \rightleftharpoons 2[AlCl_4]^-$$

$$2[AlCl_4]^- \rightleftharpoons [Al_2Cl_7]^- + Cl^-$$

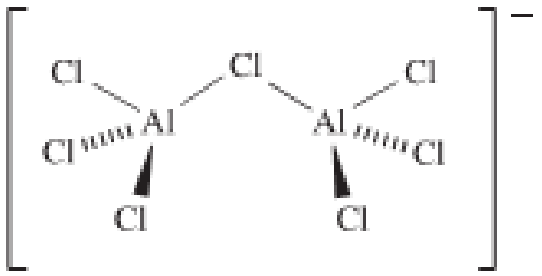
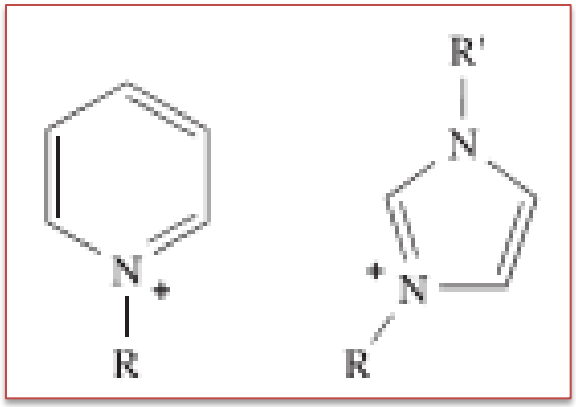


Table 9.10 Examples of ionic liquids prepared from reactions of [X]Cl and a Lewis acid where $[X]^+$ is an alkylpyridinium or dialkylimidazolium ion.

Reagents for ionic liquid formation	Anions present in the ionic liquid
$[X]Cl + AlCl_3$	Cl^-, $[AlCl_4]^-$, $[Al_2Cl_7]^-$, $[Al_3Cl_{10}]^-$
$[X]Cl + BCl_3$	Cl^-, $[BCl_4]^-$
$[X]Cl + AlEtCl_2$	$[AlEtCl_3]^-$, $[Al_2Et_2Cl_5]^-$
$[X]Cl + CuCl$	$[CuCl_2]^-$, $[Cu_2Cl_3]^-$, $[Cu_3Cl_4]^-$
$[X]Cl + FeCl_3$	$[FeCl_4]^-$, $[Fe_2Cl_7]^-$
$[X]Cl + SnCl_2$	$[SnCl_3]^-$, $[Sn_2Cl_5]^-$

alkylpyridinium ions, dialkylimidazolium ions



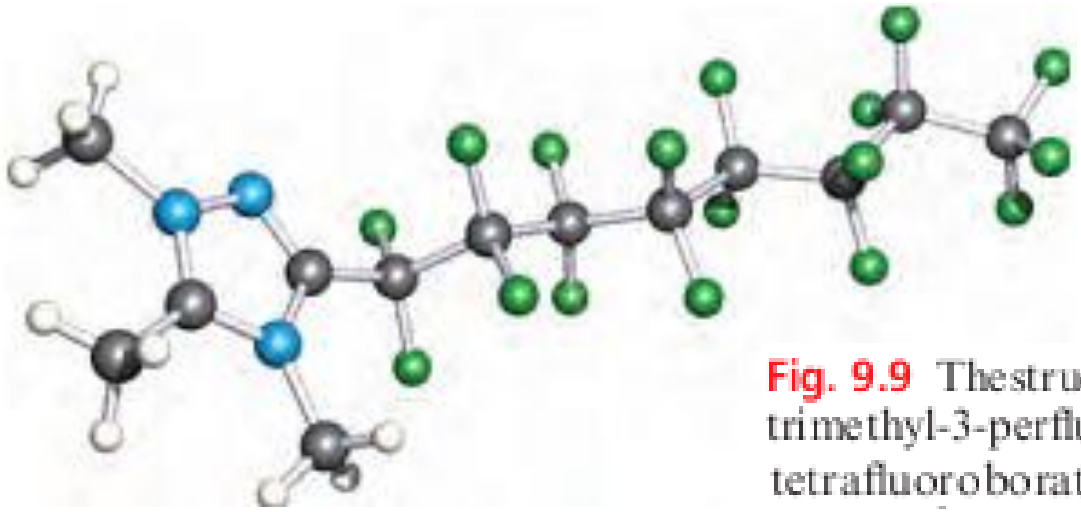


Fig. 9.9 The structure (X-ray diffraction) of the 1,4,5-trimethyl-3-perfluorooctyl-1,2,4-triazolium cation from the tetrafluoroborate salt

Colour code: C, grey; N, blue; F, green; H, white.

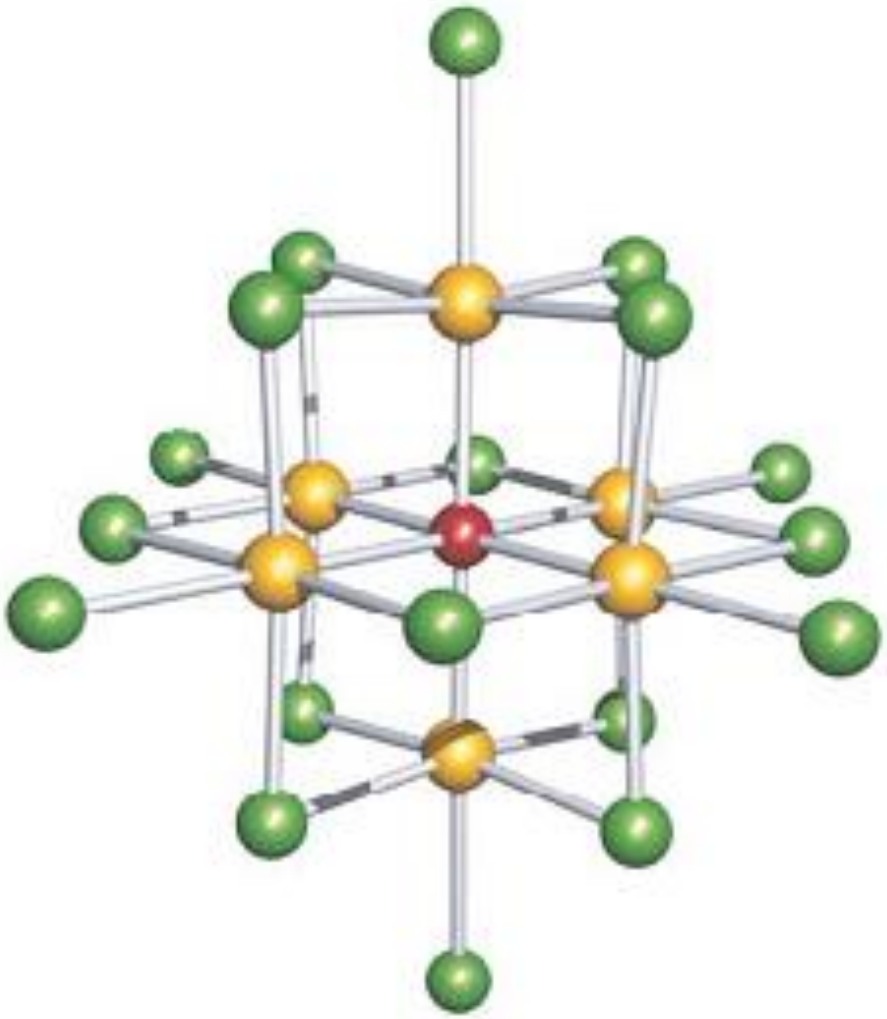


Fig. 9.10 The structure (X-ray diffraction) of the $[Zr_6MnCl_{18}]^{5-}$ anion from the 1-methyl-3-ethylimidazolium salt

Colour code: Zr, yellow; Mn, red; Cl, green.

9.13 Supercritical fluids

Properties of supercritical fluids and their uses as solvents

A supercritical fluid possesses solvent properties that resemble those of a liquid, but also exhibits gas-like transport properties. Thus, not only can a supercritical fluid dissolve solutes, but it is also miscible with ordinary gases and can penetrate pores in solids.

Supercritical fluids exhibit :

- lower viscosities
- higher diffusion coefficients than liquids.

The density of a supercritical fluid increases as the pressure increases, and as the density increases, the solubility of a solute in the supercritical fluid increases dramatically.

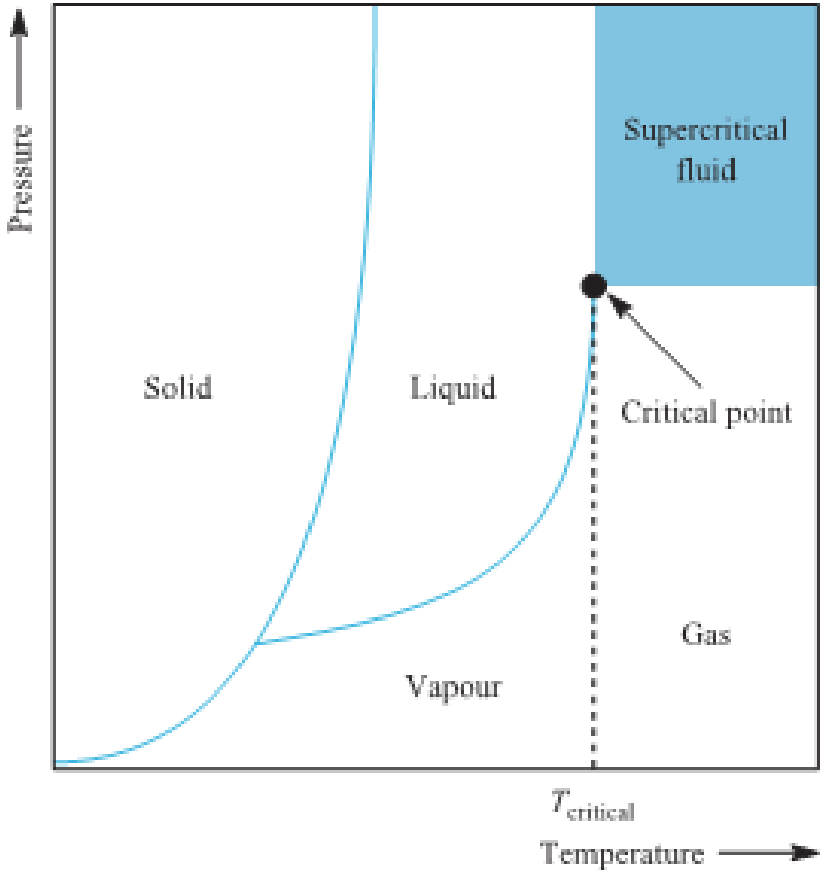
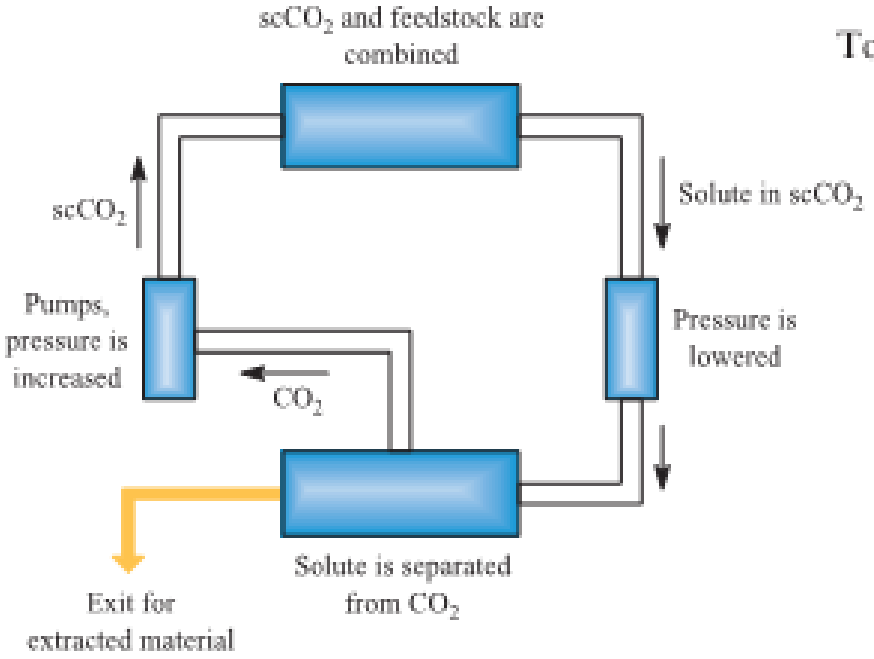


Table 9.11 lists the critical temperatures and pressures of selected compounds that are used as supercritical fluids.

Table 9.11 Critical temperatures and pressures of selected compounds with applications as supercritical fluids.

Compound	Critical temperature / K	Critical pressure / MPa†
Xenon	289.8	5.12
Carbon dioxide	304.2	7.38
Ethane	305.4	4.88
Propane	369.8	4.25
Ammonia	405.6	11.28
Pentane	469.7	3.37
Ethanol	516.2	6.38
Toluene	591.8	4.11
1,2-Ethanediamine	593.0	6.27
Water	647.3	22.05

To convert to bar, multiply by 10.



Box 9.4
Clean technology with supercritical CO_2 ($scCO_2$)

Supercritical fluids as media for inorganic chemistry

$$M(NO_3)_{2x} + 2xH_2O \xrightarrow{scH_2O} M(OH)_{2x}(s) + 2xHNO_3$$

Hydrolysis

$$M(OH)_{2x}(s) \xrightarrow{scH_2O} MO_x(s) + xH_2O$$ *Dehydration*

$$K_2S_4 + 6Ag \xrightarrow{scNH_3} K_2Ag_6S_4$$

A *homoleptic complex* is of the type $[ML_x]^{n+}$ where all the ligands are identical. In a *heteroleptic complex* , the ligands attached to the metal ion are not all identical.

$$16Fe + 128Cu + 24Sb_2S_3 + 17S_8 \xrightarrow{scNH_3} 16[Fe(NH_3)_6][Cu_8Sb_3S_{13}]$$

$$La + Cu + S_8 \xrightarrow{scNH_3} [La(NH_3)_8][Cu(S_4)_2]$$

$$Yb + Ag + S_8 \xrightarrow{scNH_3} [Yb(NH_3)_9][Ag(S_4)_2]$$

Large electronegativity difference between H (χ^P = 2.2) and F (χ^P = 4.0) results in the presence of extensive intermolecular hydrogen bonding in the liquid.

- Hydrogen bonded molecules (~7 molecules on average) in liquid phase
- Cyclic (HF)x species are present in the gas phase.

Acid- base behaviour in liquid HF

a species that produces $[H_2F]^+$ ions in liquid HF is an acid, and one that produces $[HF_2]^-$ is a base

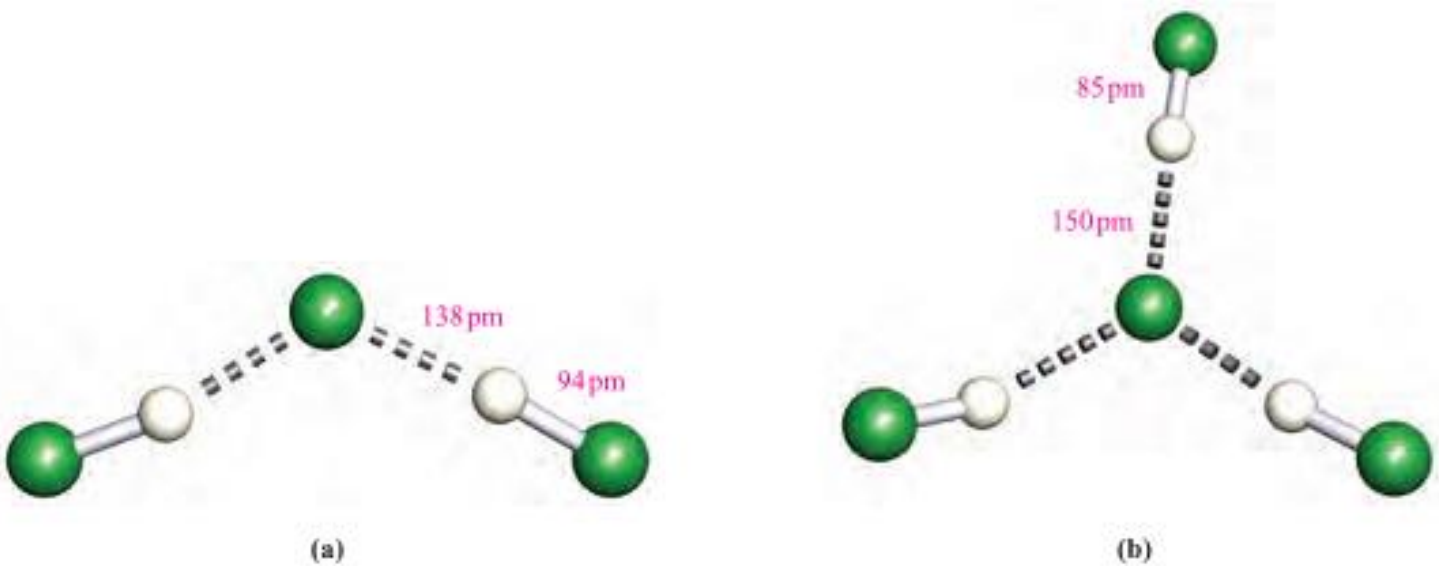


Fig. 9.4 The structures of the anions (a) $[H_2F_3]^-$ and (b) $[H_3F_4]^-$, determined by low-temperature X-ray diffraction for the $[Me_4N]^+$ salts. The distances given are the average values for like internuclear separations;